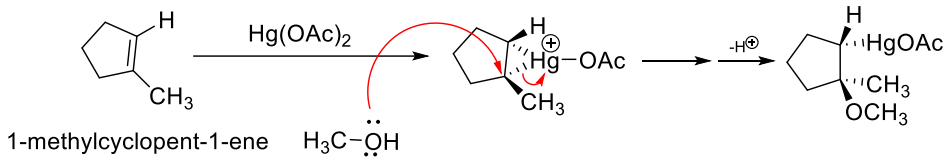
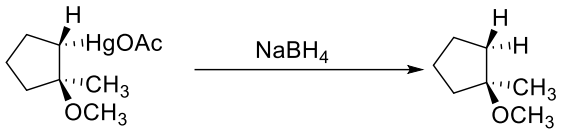
Class 3_CH2204_Sample Solution

1. Show the intermediates and products that result from alkoxymercuration–demercuration of 1-methylcyclopentene, using methanol as the solvent.

Mercuric acetate adds to 1-methylcyclopentene to give the cyclic mercurinium ion. This ion has a considerable amount of positive charge on the more substituted tertiary carbon atom. Methanol attacks this carbon from the opposite side, leading to anti addition: The reagents (HgOAc and OCH_3) have added to opposite faces of the double bond.



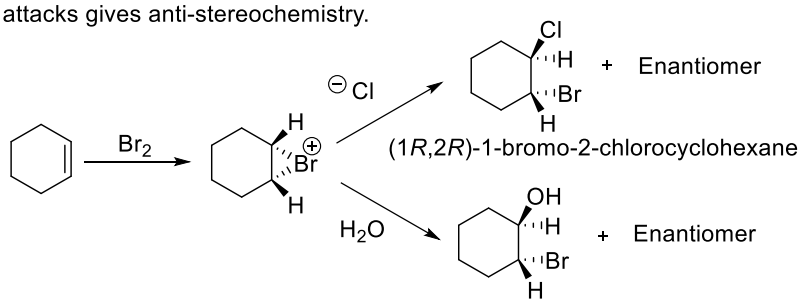
Reduction of the intermediate gives the Markovnikov product.



1-methoxy-1-methylcyclopentane

2. When cyclohexene is treated with bromine in saturated aqueous sodium chloride, a mixture of trans-2-bromocyclohexanol and trans-1-bromo-2-chlorocyclohexane results. Propose a mechanism to account for these two products.

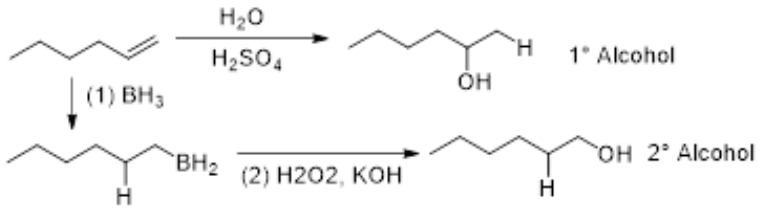
Cyclohexene reacts with bromine to give a bromonium ion, which will react with any available nucleophile. The most abundant nucleophiles in saturated aqueous sodium chloride solution are water and chloride ions. Attack by water gives the bromohydrin, and attack by chloride gives the dihalide. Either of these attacks gives anti-stereochemistry.



(1*R*,2*R*)-2-bromocyclohexan-1-ol

3. Draw the product(s) formed when $CH_3CH_2CH_2CH_2CH{=}CH_2$ is treated separately with (a) H_2O/H_2SO_4 and (b) BH_3 then H_2O_2, KOH. Comment on the product(s) obtained in both the cases.

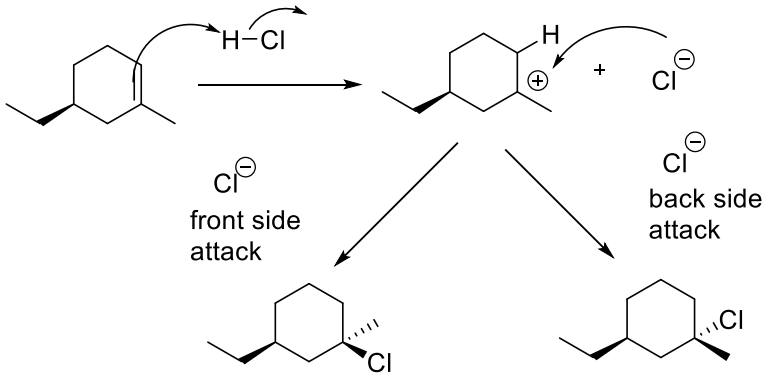
With $H_2O+H_2SO_4$, electrophilic addition of H and OH places the H atom on the less substituted carbon of the alkene to yield a 2° alcohol. In contrast, addition of BH_3 gives an alkylborane with the BH_2 group on the less substituted terminal carbon of the alkene. Oxidation replaces BH_2 by OH to yield a 1° alcohol.



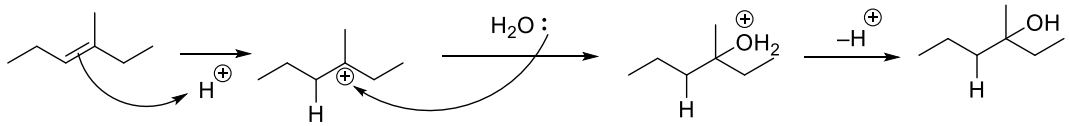
Class 3_CH2204_Sample Solution

4. Predict the product wuth stereochemistry

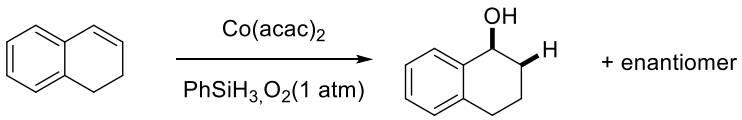
4.a.



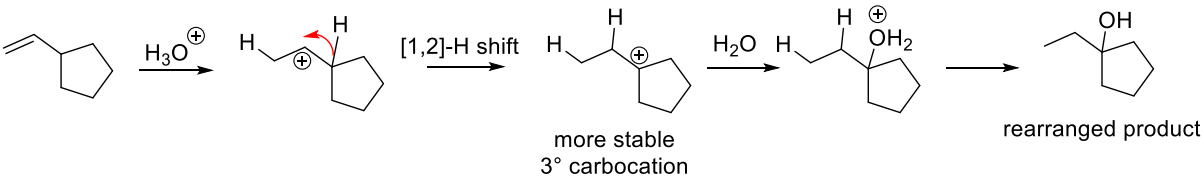
b)



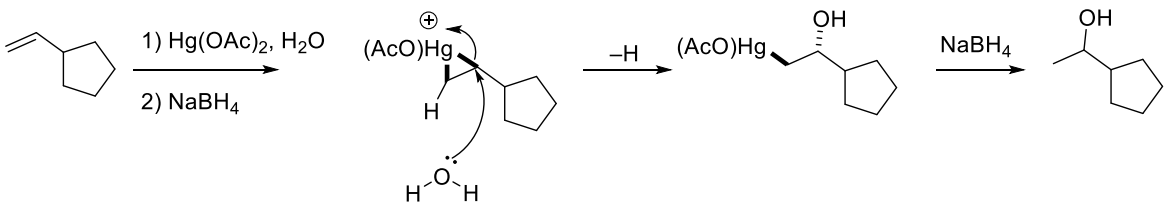
c)



d)



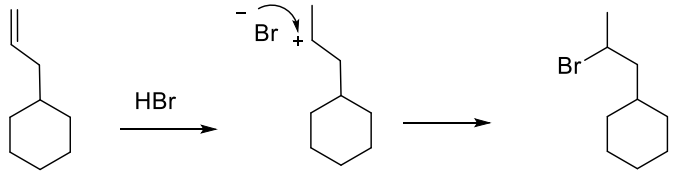
e)



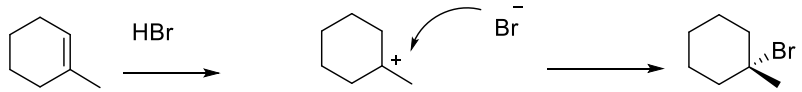
Class 3_CH2204_Sample Solution

5. Write the product for following HBr addition reactions

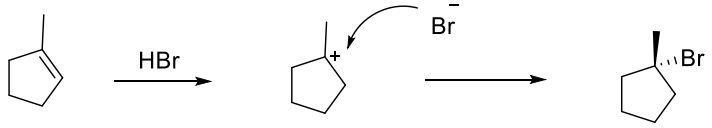
i



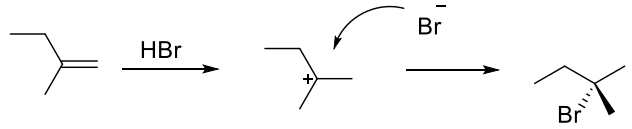
ii



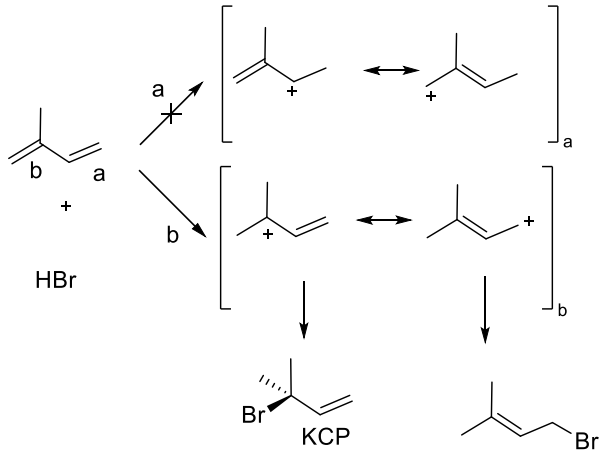
iii

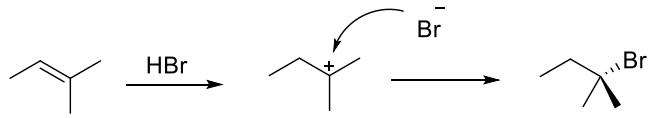


iv



v



More stable carbocation hence path b is followed

vi